

Nectar Intelligent Facilities Platform

Data Scientist Challenge — Technical Report

Reproducible report generated from current artifacts; synthetic data used because no challenge dataset was attached.

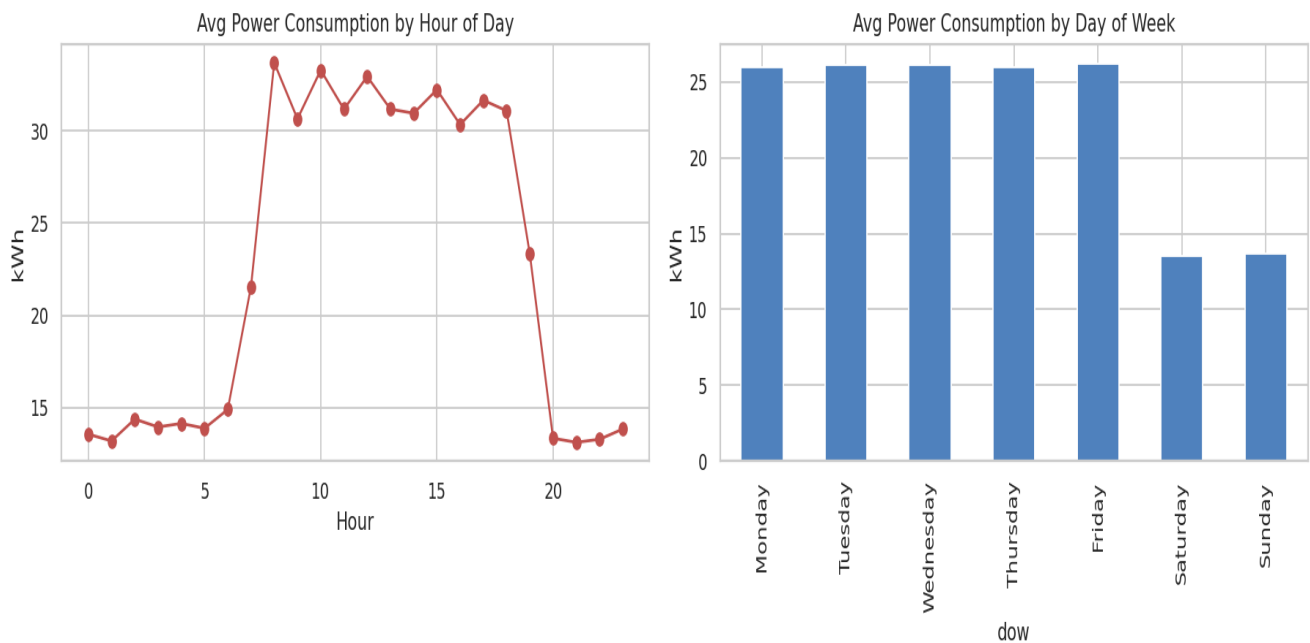
1. Problem Understanding

The solution converts connected-building telemetry into proactive maintenance, energy forecasting, anomaly detection and multi-asset impact analysis. The generated fleet contains 56 assets across 3 sites and 7 buildings and 60,453 telemetry rows with realistic IoT data-quality issues.

2. EDA

The raw audit identifies 48 duplicate readings, irregular sampling, communication gaps and about 1.3% missingness in key sensors. Chillers have the highest energy draw and business-hours energy is about 2.57x the trough.

Figure 2 — Temporal Patterns in Energy Use

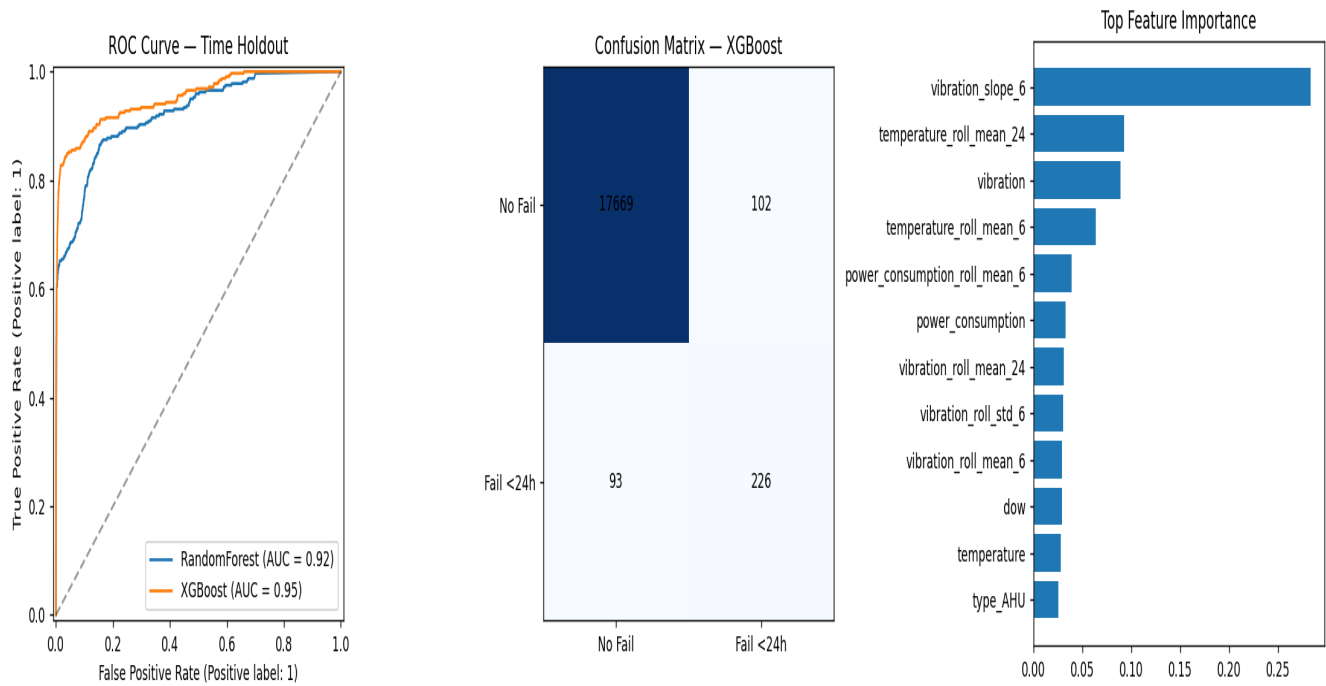


3. Predictive Maintenance

The label is the next observed fault timestamp within 24 clock hours. Active-fault rows are excluded. Features are causal and timestamp-aware; no bfill is used. A chronological split with a 24-hour embargo protects the test boundary. The alert threshold is tuned only inside training.

Model	Precision	Recall	F1	ROC-AUC
RandomForest	0.790	0.602	0.683	0.921
XGBoost	0.689	0.708	0.699	0.952

Selected model: XGBoost. Test F1=0.699, ROC-AUC=0.952, precision=68.9%, recall=70.8%.



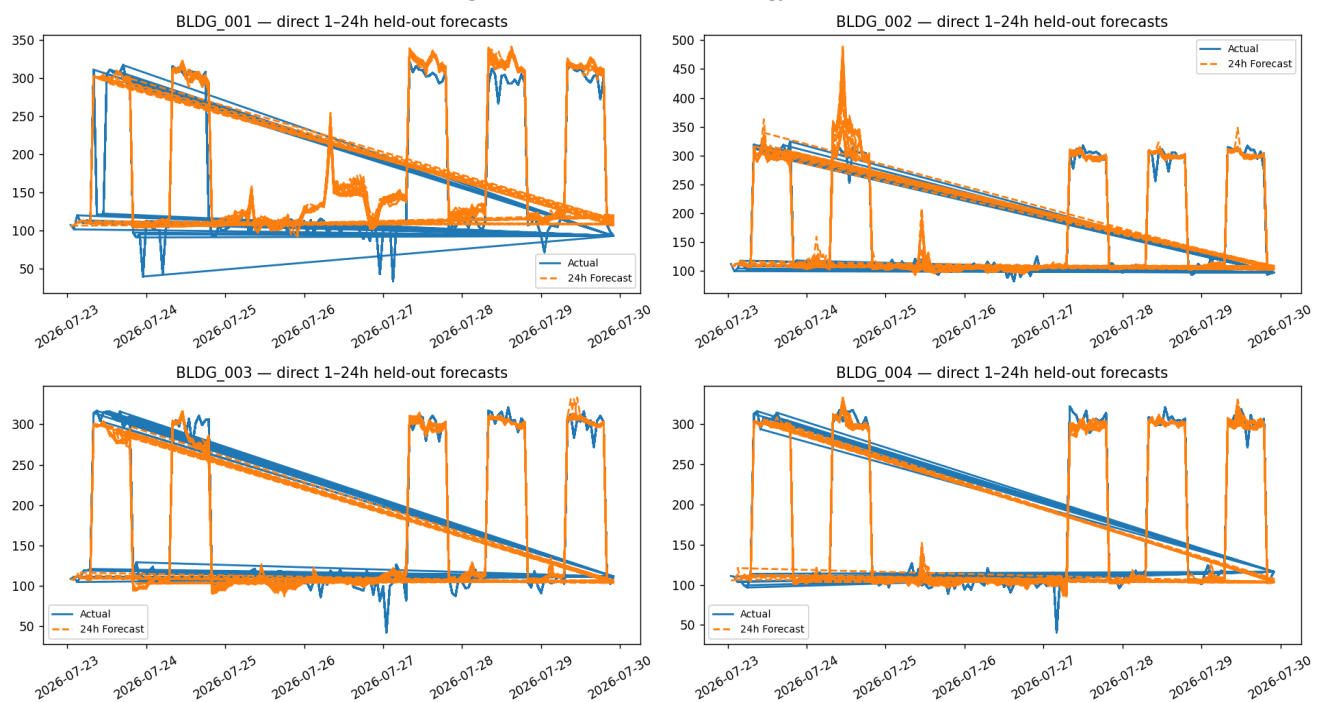
4. Energy Forecasting

A direct multi-horizon model predicts each of the next 24 hourly values from one forecast origin. The final seven days of origins are held out. No future target interpolation is used.

Metric	Model	Naive
MAE	11.57	41.68
RMSE	21.00	82.88
MAPE	9.34%	—

MAE improvement over the naive baseline: 72.2%.

Figure 8 — Direct Multi-Horizon Energy Forecasts

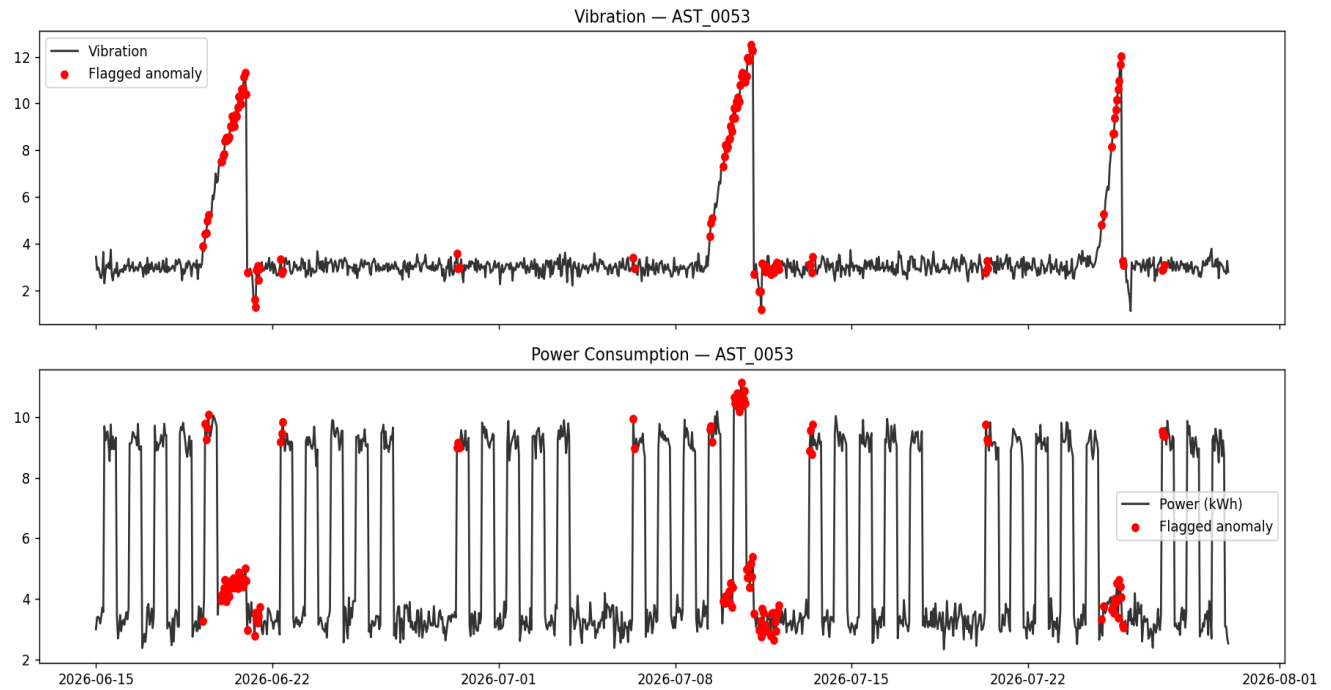


5. Anomaly Detection

Method	Flagged
Isolation Forest	607
Rolling z-score	1210
Drift	588
Union	2172

Union: 2172 readings. 1.2% overlap labeled faults; this is not treated as anomaly-detection accuracy.

Figure 10 — Anomaly Detection Example (Isolation Forest + Statistical + Drift)

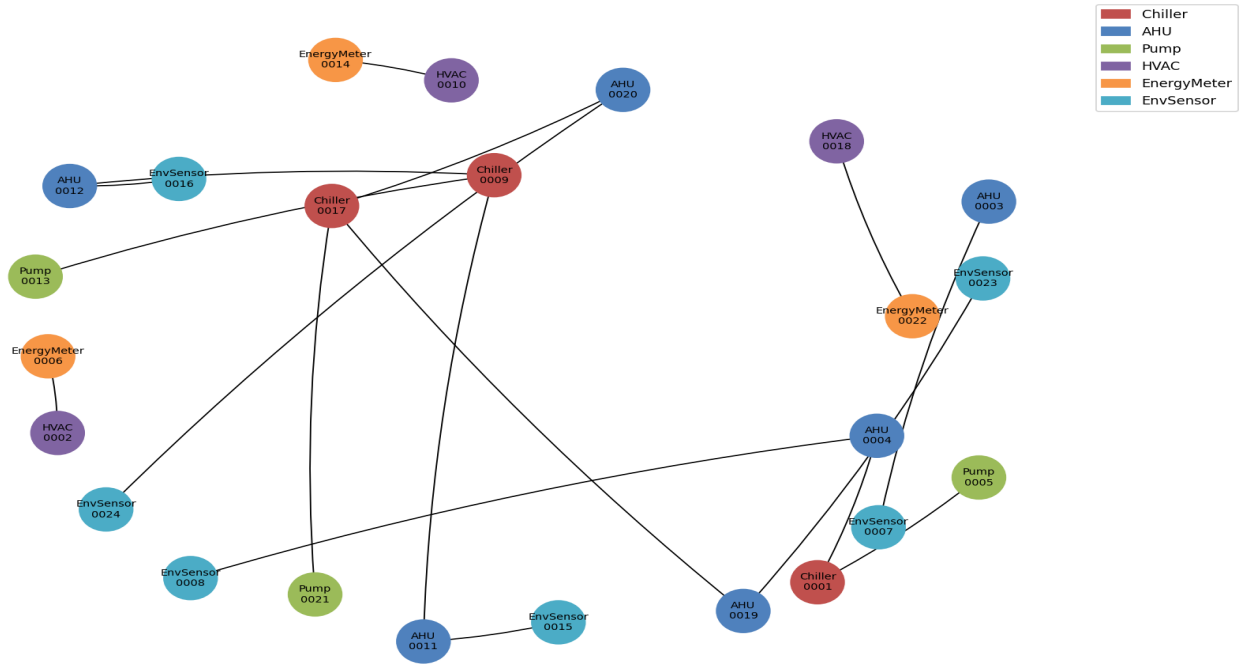


6. Connectivity

A directed graph combines hierarchy and explicit edges. The analysis is downstream impact/reachability, not probabilistic failure propagation. AST_0001 failure impacts 3 downstream assets in the example topology.

Issue	Count
Duplicate edges	1
Invalid parents	1
Orphans	1
Missing relationships	1

Figure 12 — Asset Connectivity Graph: SITE_A



7. Deployment & Next Steps

FastAPI and Streamlit now reuse the same causal feature logic and persisted preprocessing statistics. Repository-relative paths make the package portable. Next steps are real-telemetry validation, business-cost threshold tuning, technician-labeled anomaly calibration, and optional probabilistic graph propagation.